# Technical Document #755: Natural Language Processing - Comprehensive Overview

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Question answering systems extract answers from a given context. They typically consist of a retriever component and a reader component.

Tokenization is the process of breaking text into individual units called tokens. Subword tokenization methods like BPE handle out-of-vocabulary words gracefully.

Word embeddings represent words as dense vectors in continuous space. Word2Vec, GloVe, and FastText are popular word embedding techniques.

Machine translation converts text from one language to another. Neural machine translation using encoder-decoder architectures has achieved human-level performance.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

### Key Takeaways:

- Text summarization generates a concise summary of a longer text.
- Machine translation converts text from one language to another.
- Named entity recognition (NER) identifies and classifies named entities in text into predefined categories such as person nam